

Theo Bourdais

✉ tbourdai@caltech.edu

🌐 [linkedin.com/theobourdais](https://www.linkedin.com/theobourdais)

theobourdais.github.io

Education

California Institute of Technology <i>PhD in Computational + Mathematical Sciences, advised by Houman Owhadi.</i>	2022 - Present Pasadena, CA
University of Cambridge <i>Master of Mathematics (Part III), specializing in statistics. Graduated with distinction (highest grade).</i>	2020 - 2021 Cambridge, UK
Ecole polytechnique <i>Master of Science, specializing in Applied Maths and Theoretical Physics. Graduated top 10% of the cohort.</i>	2017 - 2021 Palaiseau, France

Experience

Doc.ai <i>Junior Data Engineer</i>	2021 - 2022 Palo Alto, CA
- Designed and optimized a TensorFlow training pipeline processing 20M+ images on Google Cloud VMs, leveraging ResNet50 architecture, significantly enhancing precision in computer vision tasks such as eye feature detection. - Facilitated deployment of machine learning models into production within a Go-based framework, delivering stable AI-driven services to over 100 million users. - Built medical AI systems compliant with HIPAA standards, enabling precise symptom tracking and quantification, directly supporting clinical research.	
INRIA <i>Research Intern</i>	2020 Saclay, France
- Developed novel robust optimization methods through quantile learning. - Established mathematical proofs ensuring theoretical guarantees for robustness. - Implemented and validated algorithms, demonstrating significant performance improvements in synthetic-data experiments.	

Research Interests

My research focuses on machine learning and artificial intelligence methodologies, particularly for automating scientific discovery, optimization, and predictive modeling. Specific areas include:

- **Algorithm discovery:** Developing frameworks leveraging reinforcement learning and transformer models to automate the creation of novel algorithms, enabling breakthroughs in optimization and quantum computing.
- **Data-driven Equation and Structure Discovery:** Designing flexible computational methods (using JAX, PyTorch) for automated extraction of meaningful equations and graph structures directly from large-scale data.
- **Digital Twins and Predictive Modeling:** Creating robust machine learning frameworks that enable accurate digital twins, accelerating advancements in scientific research and industry.

Projects

Data-driven equation and graph structure discovery [1] 2024	Caltech
- Developed a flexible framework in JAX to discover equations and graph structures directly from data with minimal assumptions. - Improved computational complexity from exponential to polynomial, handling up to 1,000 variables efficiently. - Released open-source implementation on GitHub and PyPI - Framework adopted in ongoing collaborations with NASA lab JPL and Sandia national Lab.	
Algorithm Discovery via Reinforcement Learning [5] 2024-2025	Caltech
- Developed a novel Reinforcement Learning (RL)-based framework for the automatic discovery of algorithms for a wide range of computational problems. - Conducted extensive computational experiments, outperforming traditional optimization algorithms on the Quadratic Assignment Problem (QAP) a challenging NP-hard problem.	

- Identified previously unknown algorithmic formulations of quantum algorithms, showcasing the method's generality and potential to advance algorithm design.
- Project funded by **DARPA**; aims to establish algorithm discovery as a foundational AI capability, with broad implications across scientific computing and beyond.

Model aggregation [3] | 2023-2024

Caltech

- Built a general ML-driven aggregation framework combining multiple models to enhance accuracy
- Improved accuracy of PDE solver by **10x**, using a Neural Operator
- Successfully applied Machine Learning techniques to scientific computing

Pruning of Neural Networks [4] | 2023-2024

Caltech

- Applied Random Matrix Theory (RMT) to prune large-scale computer vision models (ResNets, ViTs).
- Achieved parameter reduction of **40%** while maintaining accuracy within **1%** on ImageNet using **PyTorch**.

Keypoint detection for a Myasthenia Gravis study [2] | 2022

Doc.ai

- Developed advanced training strategies in **TensorFlow** for a large-scale vision model designed for eye-keypoint detection.
- Managed and optimized a pipeline handling **20M+ images**, significantly improving model performance.
- Designed and annotated specialized datasets for Myasthenia Gravis clinical research, supporting medical research.

Medication label reading | 2022

Doc.ai

- Engineered robust and maintainable production-level software in **Go** to perform medication-label recognition tasks.
- Collaborated within a 15-person team, leading to successful feature integration.
- Delivered the feature to production, serving over **100 million users** via ShareCare's healthcare app.

Publications and Pre-Prints

- [1] **Bourdais, T.**, et al. "Codiscovering graphical structure and functional relationships within data: A Gaussian Process framework for connecting the dots," In Proceedings of the National Academy of Sciences.
- [2] Lootus, M., ..., **Bourdais, T.**, et al. "Development and assessment of an artificial intelligence-based tool for ptosis measurement in adult myasthenia gravis patients using selfie video clips recorded on smartphones," in Digital Biomarkers.
- [3] **Bourdais, T.**, and Owhadi, H. "Model aggregation: minimizing empirical variance outperforms minimizing empirical error" ICLR 2025.
- [4] Berlyand, L., **Bourdais, T.**, et al. "Pruning Deep Neural Networks via a Combination of the Marchenko-Pastur Distribution and Regularization" in Arxiv 2025
- [5] **Bourdais, T.**, et al. "Discovering Algorithms with Computational Language Processing." in Arxiv 2025

Teaching and outreach

SIAM student chapter president | 2024-2025

Caltech

- Organised student talks and events for the applied Mathematics department

Reviewer

- ICLR 2024, 2025, 2026
- Journal of Computational Physics

Teaching Assistant | Winter 2024

Caltech

- Stochastic processes and regression, taught by Houman Owhadi

Instructor | March 2024

Nvidia Deep Learning Institute

- Fundamentals of Deep Learning

Distinctions and Awards

- **Kortschak Fellowship**: multi-year fellowship providing full tuition and stipend for graduate study at Caltech.
- **Ranked 1st nationwide (France)** in highly competitive entrance examinations for École Polytechnique.

Technical Skills

Programming Languages: Python, C++ (intermediate), Java (intermediate), Go (intermediate), SQL (intermediate), LaTeX.

Python packages: Jax, TensorFlow, PyTorch, Numpy, Sklearn, Pandas.

Statistical Tools: Kernel methods, neural networks, support vector machines (SVM), K-means clustering, gradient boosting

Algorithms: Classical algorithms (sorting, binary search), fundamental data structures (linked lists, hash maps), randomized algorithms.

Languages: French, English, Spanish (B2 - upper intermediate)